



Class Notes Check

Complete the warm-up problems, working with the people at your table.

TAs will be coming around to check your notes from last night.

- You will receive a stamp for each completed page
- Incomplete pages will be highlighted

Papers will not be stamped after 20 minutes after class starts.



Warm up:

pg 131
Problems:
#3.010
#3.011
#3.012

Class Notes Check:

Lesson 7A, pg 41
Lesson 7B, pg 42
Lesson 7C, pg 43
Lesson 7D, pg 44



Class Notes Check

Once your pages are checked, submit both your notes and warm-up page to Gradescope.

- Go to the Math 100 D2L Brightspace page.
- Click on **Content**.
- Navigate to **Math 100 Assignments** → **Week 5**.
- Click on **Thursday** → **Class Notes Check #7**.
- Follow the submission instructions and upload pictures of your Class Notes.

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Lesson 7D, pg 44



Classroom Expectations

Take a moment to

- Make sure you have out your class notes and something to write with
- Take out your calculator
- Put away your laptop
- Headphones off, earbuds out of ears
- Place your phone face down on your desk

QR code for virtual T184 →



Warm up:

pg 131
Problems:
#3.010
#3.011
#3.012

Class Notes Check:

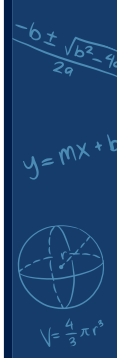
Lesson 7A, pg 41
Lesson 7B, pg 42
Lesson 7C, pg 43
Lesson 7D, pg 44

Mini Lesson

Expressions

Big Question: What can we do with an expression?

*Don't forget your notes will be graded.



Expressions Introduction

Real-life situations often include unknowns or changing quantities.

Expressions let us represent those situations with math and communicate ideas in a clear, universal language.

Not necessarily a note taking slide - just a talk through

Variables and Terms

Variable: A symbol or letter (like **x** or **y**) that represents a number which is not yet known, or might be able to change

Terms: Numbers and/or variables

Examples of Terms:

4 y
 $3x$ $\frac{x^2}{5}$

Note multiplication and division are allowed in terms

Expression

Expression: Numbers or variables joined together by addition or subtraction

Examples of Expressions:

$$3x + 4$$

$$\frac{2(y + 6)}{9}$$

Expressions from Word Problems

Turn each phrase into a two-step expression

Example:

Seven less than the product of 13 and a number

Try it yourself:

22 increased by twice Janelle's height

Go over how to choose a variable when one is not assigned

example 1: x or n

example 2: j or h

What can we do with expressions?

- **Simplify:** Take a complicated expression and make it into a simpler, *equivalent* expression.
 - Remember: What are equivalent fractions?
- **Evaluate:** Evaluate the expression by substituting numbers for variables



remember we could take 3/6 and make it 1/2

What does it mean to simplify?

For right now, to **simplify** an expression, find an equivalent expression where:

- all parenthesis are distributed $3(x + 4) \rightarrow 3x + 12$
- all like terms are combined $4x + 2x \rightarrow 6x$
- all forms of one are cancelled $\frac{6x}{6} \rightarrow x$



Note: slide transitions to help process each step

Equivalent Expressions

Definition: Expressions that look different but represent the same quantity.

Some examples are:

- 1) Distributing: $4(x + 5)$



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- 1) Distributing: $4(x + 5)$
- 2) Factoring: $3x + 12$



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- 1) Distributing: $4(x + 5)$
- 2) Factoring: $3x + 12$
- 3) Combining like terms: $8x + 12 + 3x + 10$



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Some examples are:

- 1) Distributing: $4(x + 5)$
- 2) Factoring: $3x + 12$
- 3) Combining like terms: $8x + 12 + 3x + 10$
- 4) Canceling forms of one: $\frac{4x}{4}$



Examples

Example:

Simplify the expression. Justify your steps.
 $4(5x - 3) + 5$

Try it yourself:

Simplify the expression. Justify your steps.
 $-2(5x + 3) + 5 + 2x$



Justify your steps means using the terms from the slide before (combine like terms, distribute, factor, cancel forms of one)

Evaluating Expression

To evaluate an expression:

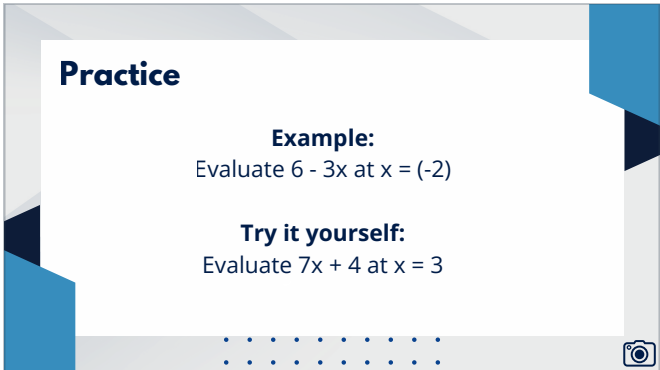
1. Plug in the number (**using parentheses**) for the variable
2. Use Order of Operations to find the value of the resulting expression



Practice

Example:
Evaluate $6 - 3x$ at $x = (-2)$

Try it yourself:
Evaluate $7x + 4$ at $x = 3$



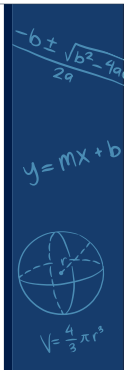
Evaluate $6 - 3x$ at $x = (-2)$

Evaluate $7x + 4$ at $x = 3$



Main Takeaways

Write out all the main takeaways that you learned from the lesson with a partner and copy down what you don't already have written in your notes.



The sidebar contains the following content from top to bottom:

- The quadratic formula:
$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
- The linear equation:
$$y = mx + b$$
- A diagram of a sphere with a radius vector r and a dashed line representing the radius.
- The volume formula for a sphere:
$$V = \frac{4}{3} \pi r^3$$

$$y = mx + b$$


In-Class Assignments

Directions:

- 1) **Rough Draft:** Work on the whiteboard (Take a picture)
- 2) **Present:** Get your work checked by a TA
- 3) **Final Draft:** Copy into Class Notes
- 4) **Submit:** Stamp + Submitted into Gradescope

Complete on Class Notes Handout



gradescope
by turnitin



- 1) **Rough Draft:** Work on the whiteboard (Take a picture)
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Gradescope Submission

Go to the Math 100 Brightspace page
(easiest on a phone)

- Nav Bar → Content
- Math 100 Assignments
- Week 5 → **Thursday** (Week 5)
- Class Assignment #7

Upload images of your paper for
each question

When you are done, work on ALEKS until the end of class



The screenshot shows the Brightspace navigation bar for The University of Arizona. The 'Content' link is highlighted with a red box. The navigation bar includes links for Course Home, Content, Grades, and Math 100 Syll.

THE UNIVERSITY OF ARIZONA MATH 100 SP26

Course Home **Content** Grades Math 100 Syll

- Nav Bar → Content
- Math 100 Assignments
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Reminders

The Midterm starts next week

- Check D2L grades for the time you signed up for
- Work in ALEKS to help you prepare.
- Math 101, bring your CatCard

Check D2L Announcements for any additional announcements

